

tion and electromagnetic interference in IoT devices.

Communication systems. Communication needs to be fast (to keep it really real-time!), secure, and reliable. High bandwidth networks are needed to ensure proper delivery of AR content, without distortion. 5G promises the high bandwidth, low latency network that the industrial metaverse needs. Ericsson is building city-scale digital twins on the Nvidia Omniverse to help accurately simulate the interplay between 5G cells and the environment for maximum performance and coverage. That's what one calls a symbiotic relationship!

According to a recent report by electronics-manufacturer, Molex, enterprises are moving to digital or software-defined circuits mainly to minimise intrusions that corrupt signals in analogue circuits. Digital circuits are not immune to noise, but they are known to be far less problematic. "To help our customers deliver stable and high-quality immersive experiences, we take extra steps in our products," says Stephen Drinan, Director of Wireless Connectivity at Molex. "For example, with our 60GHz RF chips, we incorporate re-timers into the chip, eliminating signal integrity concerns with high-definition video."

Data analytics and artificial intelligence. You collect data, fine, but what would you do with it? Once the data is aggregated, digital twins would use AI and ML algorithms to reason over that data—to find patterns, trends, and anomalies. AI can spot flaws, predict outcomes, and suggest actions. The more the data collected from the system, the better the AI system becomes—as it learns from that data. The industrial metaverse is the meeting ground of information technology and operational technologies—where data from the physical system helps improve the intelligence of the digital system; and

the insights from the digital system help improve the efficiency and efficacy of the physical system.

Extended reality. "The consumer metaverse uses VR glasses to transport people into a virtual new online world for entertainment, gaming, and shopping. The industrial metaverse, however, uses AR and MR smart glasses to make digital information and data accessible, which enriches the real world. Through intelligent software running on those smart glasses, the full potential of the industrial metaverse is unleashed. AR and MR solutions, like TeamViewer Frontline, enable digitalised workflows, blueprints, instructions, or holograms to be displayed in the right place and at the right time via the smart glasses—as if they were part of the real environment and without obstructing the workers' view," explains Witt. TeamViewer's latest add-on to Frontline, AiStudio, combines AR and AI. For example, in logistics processes, the AI can be trained to automatically verify if the correct items are picked in the warehouse. It can also be used for compliance checks or quality verification.

Cloud platforms. Software is the fabric that stitches the components of the industrial metaverse together. Microsoft offers a range of tools like Azure Digital Twins, Azure Synapse Analytics, and Microsoft Mesh with Microsoft HoloLens 2 on the cloud. Nvidia's Omniverse is another popular cloud solution for building an industrial metaverse. Siemens recently announced their new digital business platform Xcelerator. Its solutions are easy to integrate, work together seamlessly, and can be adapted very precisely to specific, individual needs. Last year, Siemens and Nvidia announced a partnership to transform the manufacturing industry with immersive experiences across the lifecycle from design through operation. Companies can

use Nvidia Omniverse and Siemens Xcelerator platforms together to enable full-fidelity digital twins and connect software-defined AI systems from edge to cloud.

Here and now

The industrial metaverse is not so far away. Quite a few enterprises around the world have started using the industrial metaverse, or at least its precursor technologies.

"I see three types of use cases to be very prominent in these initial days. First, product demonstrations for sales and marketing events and for customer experience centres. We have already implemented such a use case for a manufacturer of industrial machines. Second, customers want to enhance employee training with respect to complex processes, improve knowledge retention, and simultaneously reduce travel costs. We are presently implementing an immersive learning experience for a client that wants to train its workers in standard operating procedures and simulate specific real-world situations that would be otherwise difficult to replicate safely in classroom training. Finally, there is the digital twin. We have a history of work in this area and now we are starting to create a digital twin of a production line in the industrial metaverse," says Garg.

"Showcasing products and services in the best possible way is key to the success of any business. To tackle this challenge for a leading manufacturer in the US, we developed a mixed reality solution that creates a true-to-scale 3D hologram of heavy machinery. It allowed digital interaction between the users and the manufacturer while providing a life-like experience," says Kingsley.

Witt also recalls some of their customer success stories. Food chain Wendy's uses an AR solution running on smart glasses for livestreaming, staff training, onboarding, and